

CGV Seminar

Course organization

CS4705 Research Seminar Computer Graphics

DSAIT4215 Research in Visual Computing and Interactive Visual Analysis

2025/26

Rafael Bidarra, Elmar Eisemann, Klaus Hildebrandt, **Petr Kellnhofer**,
Ricardo Marroquim, Martin Skrodzki, Michael Weinmann

<https://graphics.tudelft.nl/people>

Relation to CS/DSAIT Themes

- **CS4705** Research Seminar Computer Graphics.
 - CS Theme Computer Graphics.
- **DSAIT4215** Research in Visual Computing and Interactive Visual Analysis.
 - DSAIT Theme Visual Data Science.
 - DSAIT Theme Visual Computing.
- Two codes -- One course.
 - The difference lies in **your paper choice**.

CGV Colloquium

- The seminar is part of **CGV Colloquium** series.
 - Invited talks on CGV topics from internal & external guests.
 - Happening throughout the year.
- **Q3:** Special sessions for (voluntary) mid-term presentations of MSc projects
 - Practice presentation skills.
 - Get feedback for your project.
 - Will be announced by e-mail to MSc students registered in MaRE.

Instructors

CGV: <https://graphics.tudelft.nl/people/>



Rafa Bidarra



Elmar Eisemann



Klaus Hildebrandt



Petr Kellnhofer



Ricardo Marroquim



Martin Skrodzki



Michael Weinmann

Course goals

- Acquire in-depth knowledge on a specific CGV topic.
Topic can but does not need to be related to your MSc thesis subject.
- Learn how to **read**, **present** and **discuss** scientific papers.

Assessment structure

50 %

**paper presentation
and discussion**

40%

**implementation
(incl. report and
presentation)**

10 %

Questions

Learning objectives (from the Study Guide)

At the end of this course, the student is able to:

1. Explain representative methods covering selected topics ... in form of a **public lecture**.
2. Formulate insightful **questions** during presentations of others.
3. Implement **a tool** that helps to further **explain** the discussed technique.
4. Critically assess results of the implementation in form of a **short report and a presentation**.

Presentation

Questions

Implementation

Implementation

1. Paper presentation: Schedule

50 %

paper presentation
and discussion

duration (min)	Activity	Comments
10	paper presentation	- everybody will think about one suggestion and one good element of the presentation form
3	Q&A discussion (on the content)	- kicked off by the 2 graded questions - all other students also ask questions as time allows
2	feedback about the presentation	- audience gives feedback on the presentation - the next speaker gets ready
15	TOTAL	

This schedule will be adjusted based on the actual number of course participants.

1. Paper presentation: Logistics

50 %

paper presentation
and discussion

- Your presentation must be **uploaded to Brightspace** by **9 am** on your presentation day.
- **It is not allowed to present from your own device.** ← We do this to streamline speaker switching.
- All presentations will be played from a staff computer in the lecture room using your submission on Brightspace.
- Available SW: **MS PowerPoint**, PDF viewer, Web browser (with internet connection).

2. Implementation: Educational Tool

40%

implementation
(incl. report and
presentation)

- The goal **is NOT** to simply reimplement the paper.
- The goal is to support **education** and further **explain** the paper.
 - E.g., visualize how it works, how it fails, the effect of parameters, analyze the study in a different context...
- The work is expected to be substantial.
 - 40% of 5 EC = 56 hours of work.
- Further details in **a separate document** on Brightspace.

2. Implementation: Plan Feedback

40%

implementation
(incl. report and
presentation)

- Formative feedback for the implementation plan will be organized during one of the contact lectures.
- Details will be announced later when the schedule is available.

Attendance

- Sessions are held in **a physical form only**.
- We expect everyone to **attend** every session in person.
- We might record the presentations for grading purposes.
 - Likely using Zoom or Teams. Inform us if that is an issue.

3. Questions

10 %

Q&A

- **At least one** mandatory question per course.
- **Randomly selected** from **ALL** enrolled students during the Q&A session.
 - Typically, two questions asked after each presentation.
 - You can only be selected once per session.
 - You can be selected in multiple sessions.
- Your question should demonstrate that you have **followed the presentation** (binary grading: **FAIL=1** or **OK=10**).
 - No general questions such as “How would you extend this?”
- **Total grade** is an average of all received Question grades.

3. Questions - Absences

10 %

Q&A

- Absence can be **informed** or **uninformed**.
 - **Informed** absence must be announced by email to p.kellnhofer@tudelft.nl before the official start of the class (timestamp decides).
 - **Only two** informed absences possible per course. Further absences count as uninformed.
- If you are absent (informed or not) you are guaranteed to be selected to ask a question which results in a **failed question** = extra grade of 1.0.

3. Replacement of failed questions

- **Up to 3 failed** questions can be **partially** replaced by 2x video presentations of 2 papers from the missed session – 5 minutes per paper.
 - This is increased to 4x video for **uninformed absences**.
- Submit to Brightspace->Assignment->"Failed question replacement" by June 28.

10 %

Q&A

3. Question replacement - Rubric

- Replacement grade for each failed question is capped:
 - 1st replacement: Max grade = 10.0.
 - 2nd replacement: Max grade = 8.0.
 - 3rd replacement: Max grade = 6.0.
 - 4th and further replacements: Not possible (Max grade = 1.0).

10 %

Q&A

3. Questions – example situations

10 %

Q&A

- Two good questions + 1x missed not replaced:
 - $\text{Average}(\{10, 10, 1\}) = 7.0 \Rightarrow \text{PASS}$
- Two good questions + 4x replaced (e.g., illness):
 - $\text{Average}(\{10, 10, 10, 8, 6, 1\}) = 7.5 \Rightarrow \text{PASS}$
- One good question + 7x replaced:
 - $\text{Average}(\{10, 10, 8, 6, 1, 1, 1, 1\}) = 4.8 \Rightarrow \text{FAIL}$
- No in-class + 8x replaced:
 - $\text{Average}(\{10, 8, 6, 1, 1, 1, 1, 1\}) \Rightarrow 3.6 \Rightarrow \text{FAIL}$

Passing criteria

- As defined in **Rules and Regulations of the board of examiners** Article 17.6.
- Must pass both:
 - Each of the 3 partial grades ≥ 5.0 (rounded to 0.1)
 - The final grade ≥ 6.0 (rounded to 0.5)

Repair options

- **Under conditions** described in TER Implementation Regulations Article 5.5.
- **Presentation:** An individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student.
- **Implementation:** A new implementation following a new approved plan.
- **Questions:** A short video presentation(s) of assigned paper(s).
- The **maximum** grade for each repair option is a **6.0**.

Brightspace

- You **must enroll** to the seminar course **TWICE** in Brightspace!
 - 1) A shared page [CGV Seminar](#): All course information except the grades.
 - 2) A course specific page [CS4705](#) - or – [DSAIT4215](#) : The grades will be published here.
- Submissions through the “**CGV Seminar**” Brightspace page:
 - Paper, presentation slides, questions, report, implementation, video presentation, etc.

Choosing a paper to present

What to present and what to implement

Option 1: Default paper selection system

- We will upload **a list of available papers** to Brightspace.
 - It will be available upon Announcement on **Wed April 22 at 12:00pm (noon)**.
- You can choose a paper by enrolling to a Brightspace group.
 - Collaboration -> Groups -> Paper Selection
 - Choose a Group ID matching the paper ID from the list.
 - First come, first served.

Option 2: Custom selection

- If you already have a good idea what you want to present, you can choose your own paper (different from those on the list).
- The choice must be **pre-approved** by a course instructor = “**Supervisor**”.
 - This assumes that you know how to identify a good paper, e.g., through a prior knowledge of the field.
 - Make sure to also consider the implementation plan!
- If your paper is not approved, **you must follow Option 1**.

Selection Process (common for both options)

- Paper selection deadline: **Wed Apr 29 at 18:00**
- [Everyone] Submit the chosen paper as a PDF to Brightspace.
 - Assignments -> **Selected paper**
- [Option 2 only] Enroll yourself to a “**Supervisor**” group
 - Groups -> Supervisors -> [name of an instructor]

Policy on use of Generative AI

- For brainstorming/idea generation, writing quality enhancement (improve readability and language of the work), and general formatting of references you are allowed to use any type of generative AI tool. **The use must be declared and explained in the report.**
- For other purposes including coding, generative AI tools may be used. The use must be declared and explained in the report in detail. It must be clear what work can be attributed to the student and what to the AI. **Only your own contribution is evaluated, not the work of AI.**
- Undeclared use of AI tools as well as incomplete or misleading explanation of use of AI tools anywhere in the course is **fraud**.

Timeline

Date	Content
Tue Apr 21	Kick-off session
Wed Apr 22, 12:00	Default paper list released
...	...
Wed Apr 29, 18:00	Paper selection deadline (submit to BS)
Fri May 1	Presentation schedule published
Mon May 4 and onwards	Presentations (slides by 9am)
...	...
Fri June 19	Implementation (submit to BS)

